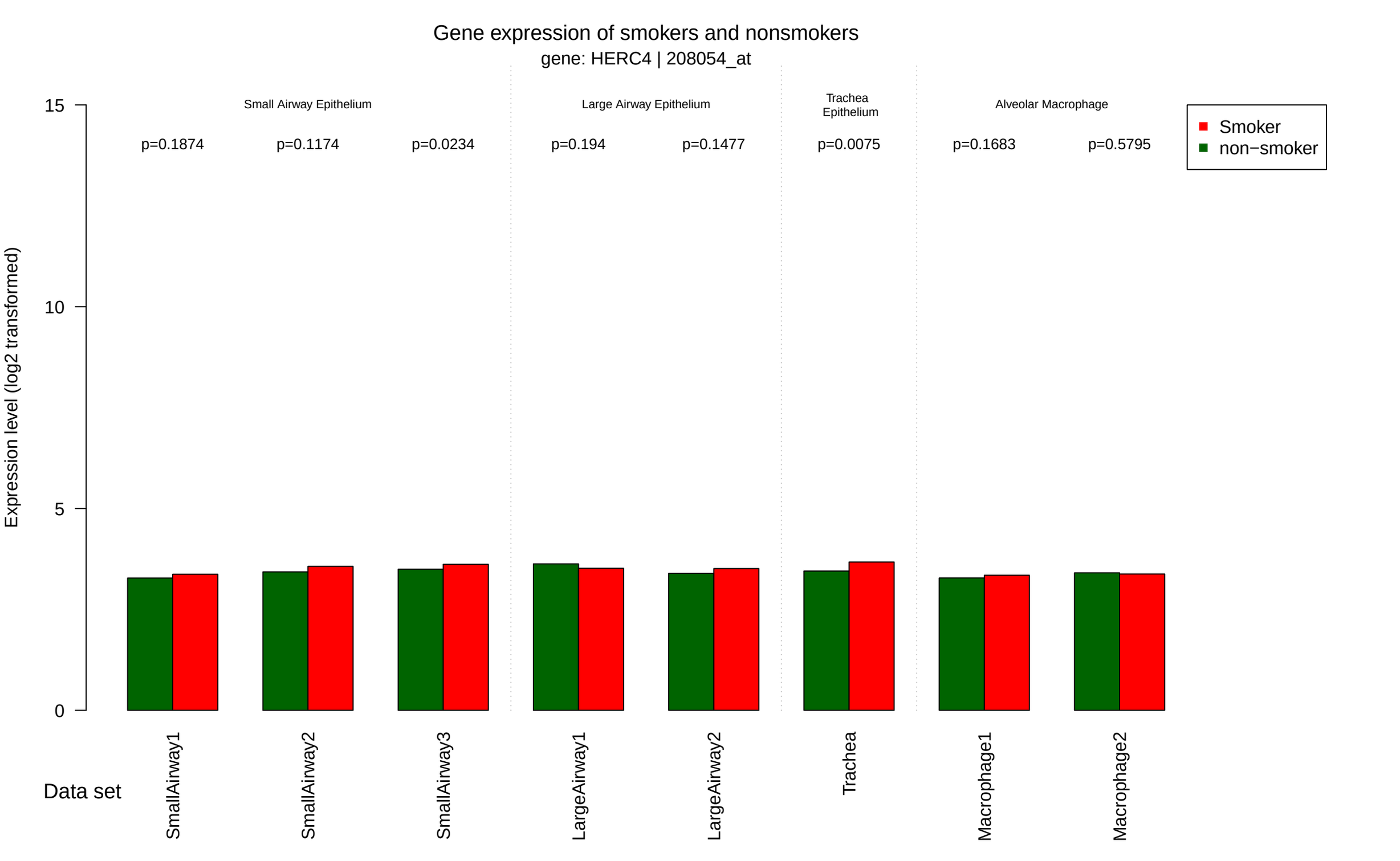
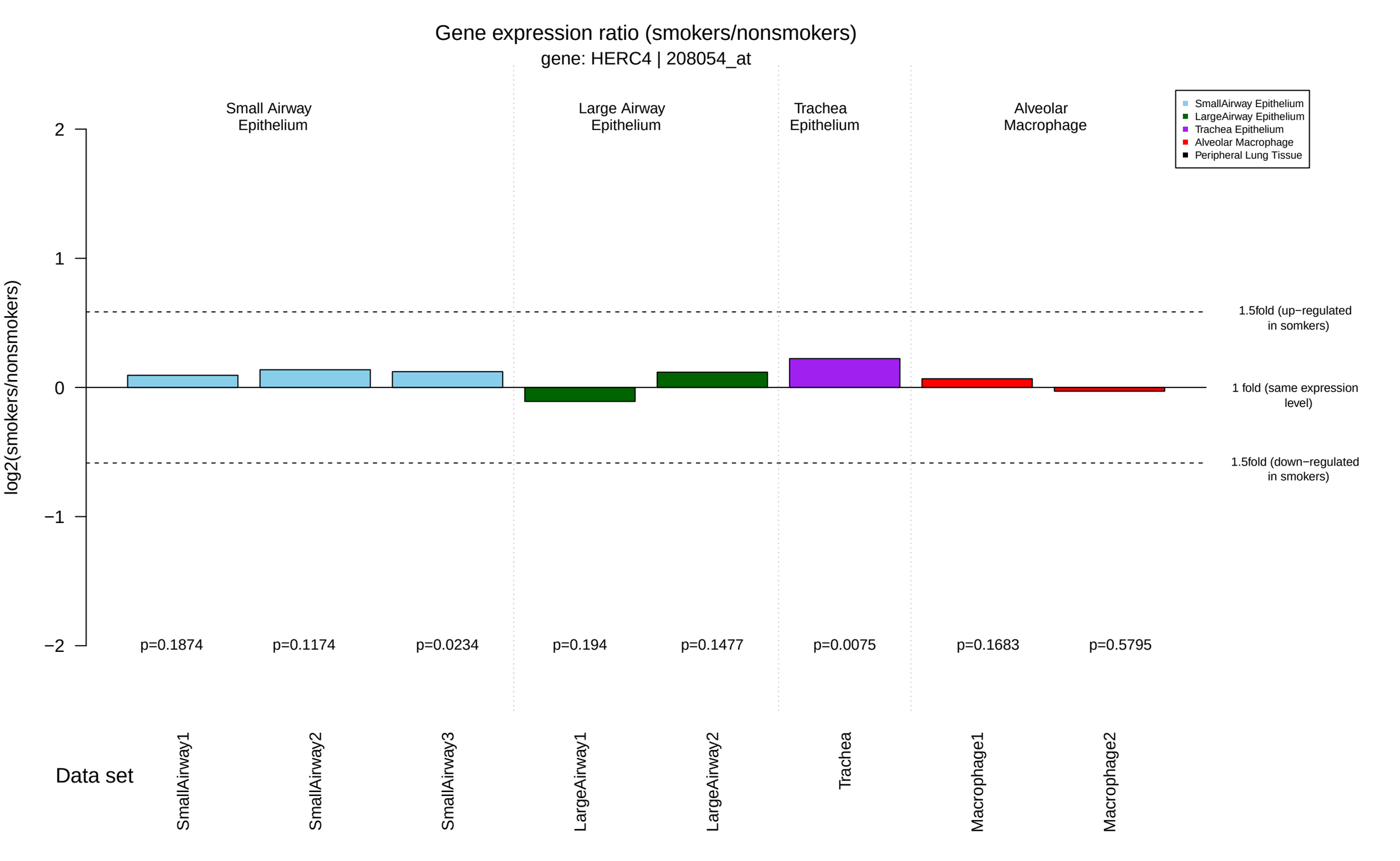
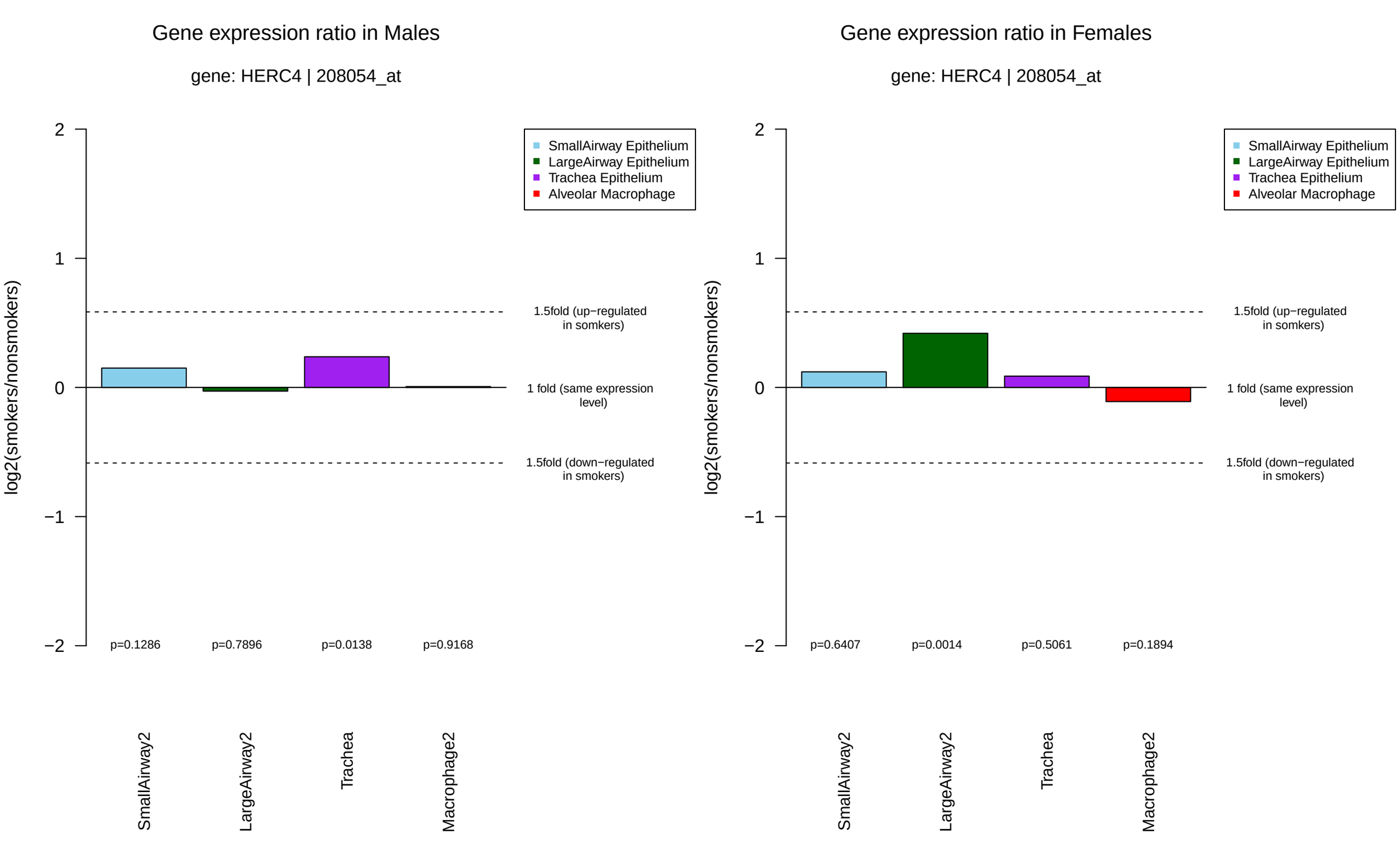
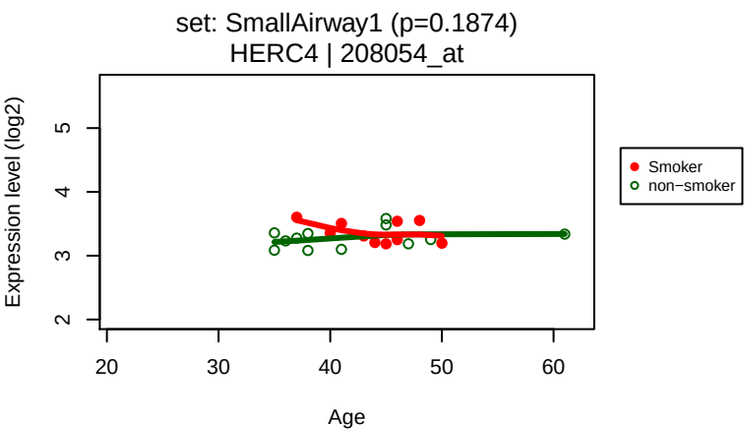
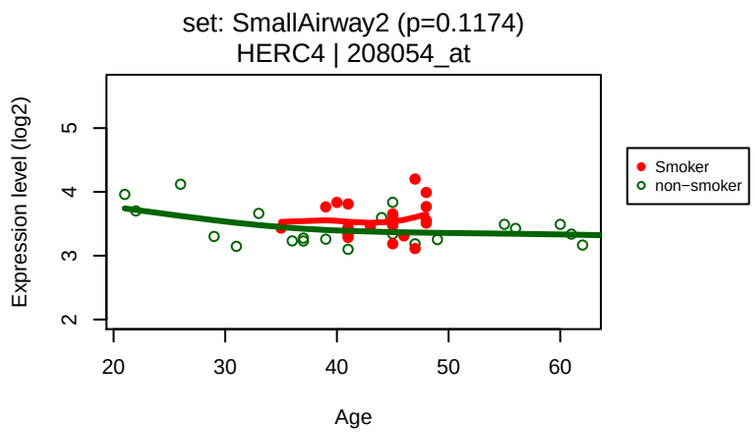
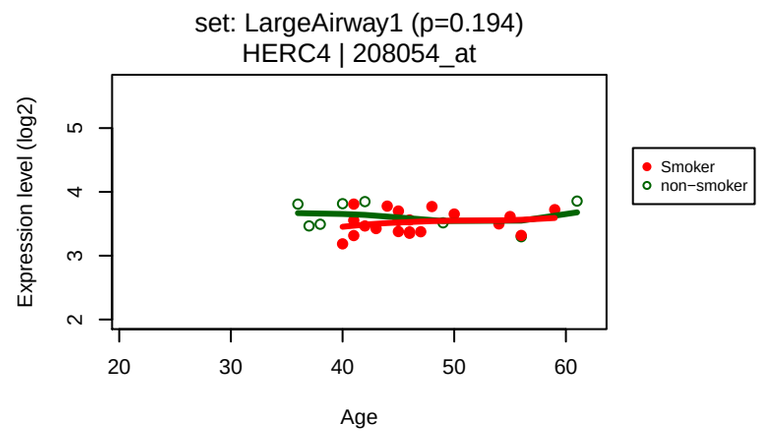

Gene expression of smokers and nonsmokers
gene: HERC4 | 208054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1874
p=0.1174
p=0.0234
p=0.194
p=0.1477
p=0.0075
p=0.1683
p=0.5795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

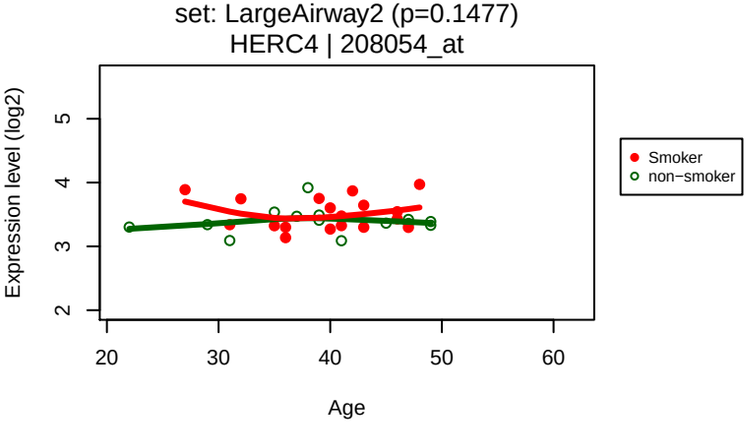

Gene expression ratio (smokers/nonsmokers)
gene: HERC4 | 208054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1874
p=0.1174
p=0.0234
p=0.194
p=0.1477
p=0.0075
p=0.1683
p=0.5795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

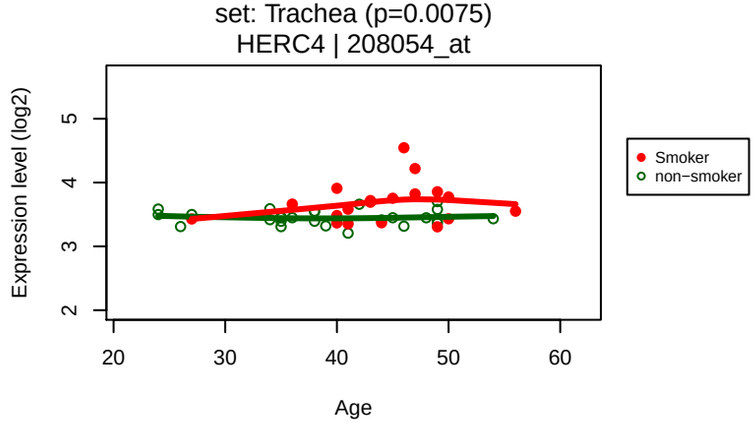

Gene expression ratio in Males
gene: HERC4 | 208054_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1286
p=0.7896
p=0.0138
p=0.9168
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HERC4 | 208054_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6407
p=0.0014
p=0.5061
p=0.1894
SmallAirway2
LargeAirway2
Trachea
Macrophage2

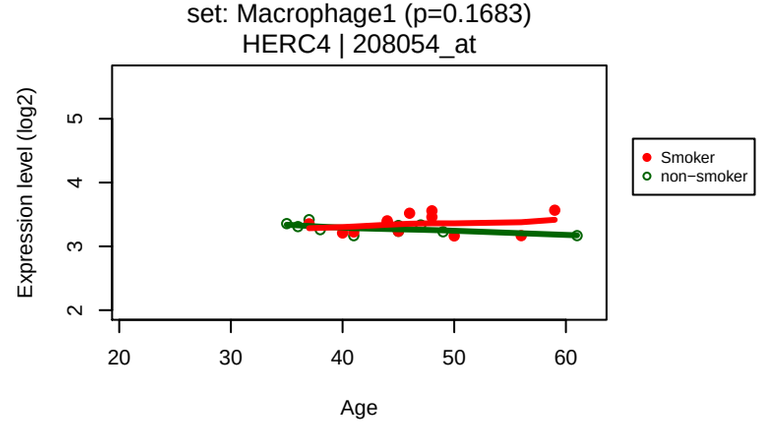

set: SmallAirway1 (p=0.1874)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker

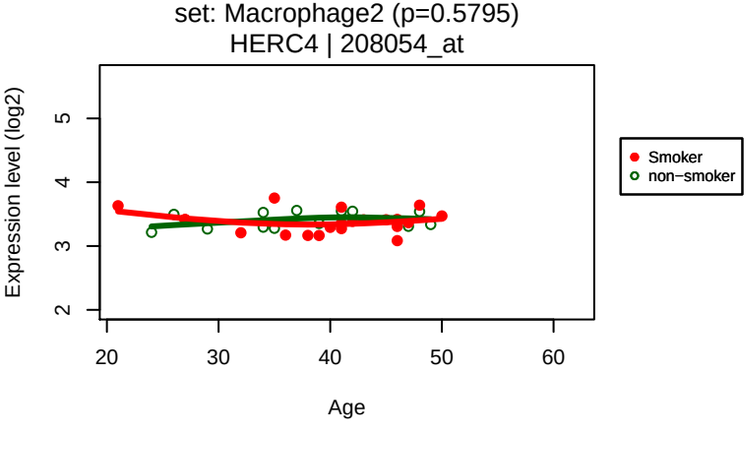

set: SmallAirway2 (p=0.1174)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker

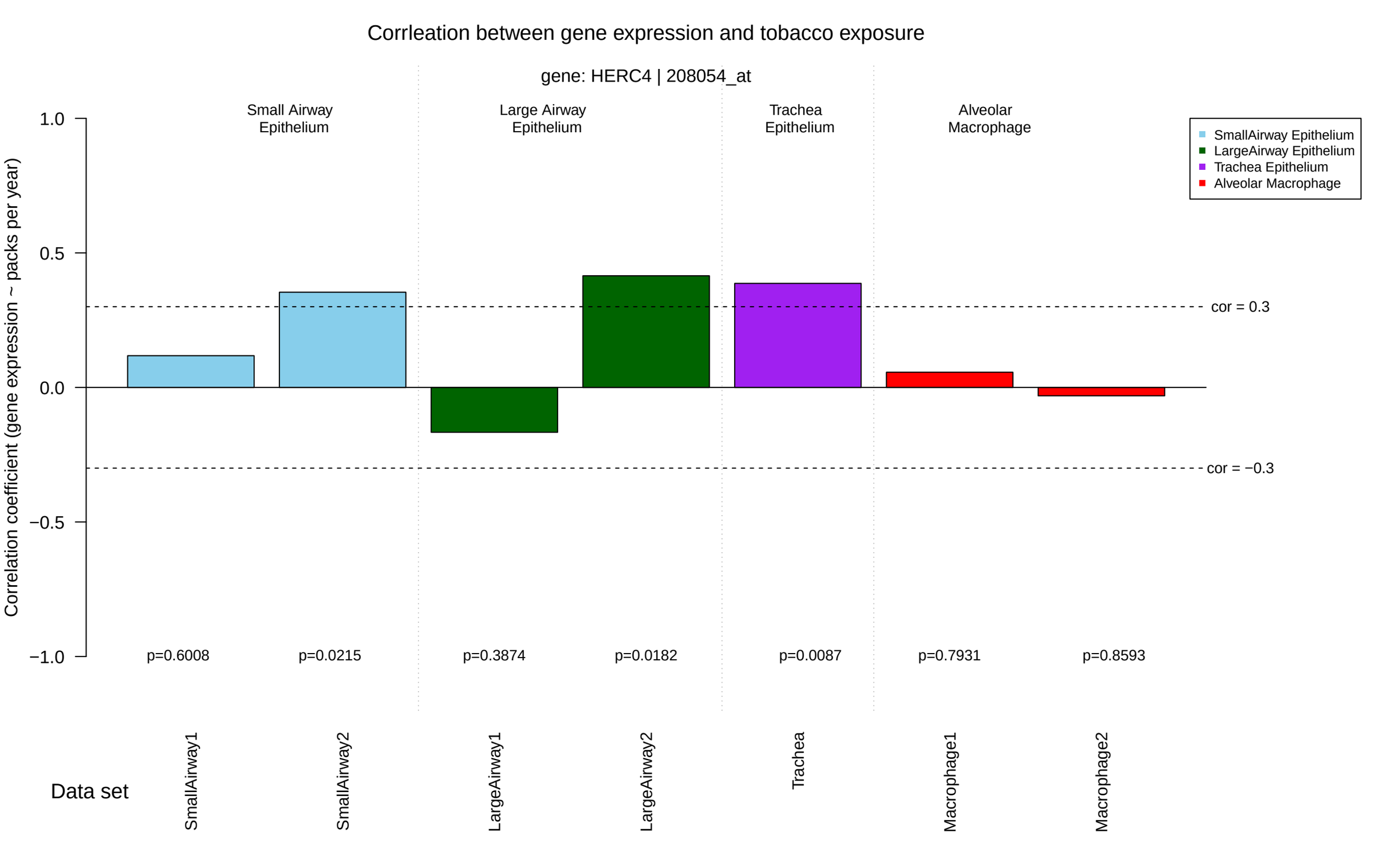

set: LargeAirway1 (p=0.194)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1477)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1683)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5795)
HERC4 | 208054_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HERC4 | 208054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6008
p=0.0215
p=0.3874
p=0.0182
p=0.0087
p=0.7931
p=0.8593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2